Yu Zhu

+86-13895127909 | 23210220065@m.fudan.edu.cn | Google Scholar

Education

- **M.Sc., Fudan University** *09 2023 - 06 2026*
  Department of Chemistry Shanghai, China
  - Supervisor: Prof. Xu Xin and Jr. Res. Sai Duan
  - Research interest: Machine Learning Force Field, Open-Set Recognition, Scanning Tunneling Microscope.
- **B.Sc., Soochow University** *09 2019 - 06 2023*
  College of Chemistry, Chemical Engineering and Materials Science Suzhou, China
  - GPA: 3.8/4.0 (92.1/100)
  - Supervisor: Prof. Jianlin Yao and Assoc. Prof. Minmin Xu
  - Research interest: Surface-enhanced Raman spectroscopy, Fingerprint Imaging.

  Note: Received verbal admission to the Ph.D. program (Fall 2026) at California Institute of Technology, advised by Prof. William A. Goddard III.

Experience

- **Microsoft Research[]** *07 2024 - 06 2025*
  AI for Science Intern Shanghai, China
  - Contributed to Microsoft's flagship materials modeling framework, **MatterSim**.
  - Developed generalized benchmarks to inform model architecture design and training data generation.
  - Inferenced on models up to **1B parameters**; managed high-throughput datasets (up to **200 M** entries); builded highly parallelizable computational pipelines on Azure Cloud handling **10 K concurrent tasks**.
  - Familiar with MatterSim and other universal machine learning force field model architecture design.

Projects

- **Reconstruct Molecular Orbitals from STM Images via Artificial Intelligence Approaches** *01 2024 - 07 2024*
  In a nutshell: Are orbitals observable? []
  - **Project significance:** Constant-height STM imaging with a metallic tip yields tunneling currents directly related to MO spatial distributions(s-wave, **Tersoff-Hamann theory**) but suffers from low resolution, whereas CO-functionalized tips afford enhanced spatial detail by introducing high-angular-momentum (p-wave, **Chen's derivative rule**) components, yet the comparable I_s and I_p contributions to the total current obscure the true MO signal, preventing direct interpretation of the images as MO maps .
  - **Responsibilities:** This is my main research project for the MS.c graduate program where I am responsible for **(1)** simulating STM images at the DFT level using Bardeen's approximation and **(2)** developed STM-Net, a physics-driven U-Net, segmenting out s-wave (MO) and p-wave (noise) contributions. **(3)** By incorporating Tersoff–Hamann theory and Chen's derivative rule to analyze how CO-functionalized tips imaging, STM-Net reaches a **PSNR > 37 dB** with only **10%** of the training data (**24 images**) and exceeds **43 dB** when trained on the full dataset, indicating near-perfect recovery of true s-wave STM maps.
  - **Key skills:** STM simulation | models(DAE, UNet, Mask2former, etc.)
- **MatterSim: A Deep Learning Atomistic Model** *07 2024 - 06 2025*
  In a nutshell: A unified multimodal foundation model for precise and generalizable in silico materials characterization. []
  - **Project significance:** MatterSim was trained on 35 million first-principles-labeled structures, enabling zero-shot prediction of key material properties across elements and extreme conditions. By learning a unified representation across physical modalities, it establishes a transferable materials "language", marking a shift from task-specific models to a generalizable framework. This significantly accelerates in silico characterization and high-throughput materials design.
  - **Responsibilities:** see **EXPERIENCE** for details
  - **Key skills:** vasp | python(atomate2, joblib, phonopy, phono3py) | models(esen, mace, orb, etc.)
- **MPBR: Multimodal Progressive Bidirectional Reasoning for OSFGR** *01 2025 - 06 2025*
  In a nutshell: Suppressing semantic drift in open-set recognition(OSR).

- **Project significance:** The purpose of this study is to explore a dynamic learning paradigm in a new open-world scenario: we first obtain an unknown distribution description from open-set learning and then, under the supervision of large language models, employ incremental learning to internalize this unknown distribution into our known knowledge base. In prior work, I designed a coarse- and fine-granularity–compatible open-set recognition model—Multimodal Progressive Bidirectional Reasoning (MPBR)—which outperforms the state of the art on the largest open-set dataset, iNat2021-OSR, by 2.6 % in AUROC and 2.5 % in OSCR; this work has been submitted to ICCV 2025.
- **Responsibilities:** Collaborated with a colleague during the internship, I am responsible for **(1)** developing benchmarks pipelines and **(2)** evaluating leading OSR models on iNat2021-OSR to inform architecture design. **(3)** Familiar with MPBR and other OSFGR model architecture design.
- **Key skills:** visualization(t-SNE, Grad-CAM) | models(swim-transformer, HAN-OSFGR, etc.)

- **SERS Imaging on Flexible Substrates for Fingerprint Identification Research** *04 2021 - 06 2023*

In a nutshell: Perhaps smoking offenders never imagined that the nicotine on their fingerprints could be used for imaging.

- **Project significance:** Developed a Surface-Enhanced Raman Scattering(SERS)-based imaging approach for fingerprint recognition. The approach enables accurate chemical mapping of both latent and live fingerprints under variable environmental conditions (e.g., surface material, temperature, humidity, aging), overcomes challenges in trace evidence detection on non-planar surfaces. Moreover, the work provides a novel route to non-invasive, label-assisted chemical identification of fingerprint residues, such as drugs or nicotine, offering new possibilities for suspect profiling.
- **Responsibilities:** Collect fingerprint samples, perform SERS imaging, and analyze the results.
- **Key skills:** SERS Mapping | visualization(Origin)

Patents and Publications

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[S.1] ***Zhu, Y.***, Xue, R., Ren, H., Chen, Y., Yan, W., Wu, B., Duan, S., Zhang, H., Chi, L., & Xu, X. (2025). **Reconstructing Pristine Molecular Orbitals from Scanning Tunneling Microscope Images via Artificial Intelligence Approaches**. *J. Am. Chem. Soc. Au*.

[S.2] Li, J., Chen, Z., Wang, Q., Yang, H., Lu, Z., Li, G., Chen, S., ***Zhu, Y.***, Liu, X., Tan, J., Tang, M., Zhou, Y., Zeni, C., Fowler, A., Zügner, D., Pinsler, R., Horton, M., Xie, T., Liu, T.-Y., Liu, H., Qin, T., Lv, B., Donadio, D., & Hao, H. (2025). **Probing the Limit of Heat Transfer in Inorganic Crystals with Deep Learning**. DOI: 10.48550/arXiv.2503.11568

[S.3] Yang, H., Liu, X., Hu, C., Zhou, Y., Shi, Y., Liu, C., Li, J., Li, G., Wang, Q.,***Zhu, Y.***, Chen, Z., Chen, S., Zeni, C., Horton, M., Pinsler, R., Fowler, A., Zügner, D., Xie, T., Smith, J., Sun, L., Chen, Y., Bai, Y., Kong, L., Hao, H., & Lu, Z. (2024). **MatterSim: A Deep Learning Atomistic Model Across Elements, Temperatures and Pressures**. *Submitted*.

[S.4] Tan, J., Jing, P., ***Zhu, Y.***, Liu, Y. (2025). **MPBR: Multimodal Progressive Bidirectional Reasoning for Open-Set Fine-Grained Recognition**. *International Conference on Computer Vision*.

[S.5] Wu, M., Tan, J., ***Zhu, Y.***, & Ren, J. (2026). **KappaFormer: Physics-aware Transformer for lattice thermal conductivity via cross-domain transfer learning**. *arXiv*. DOI: 10.48550/arXiv.2604.03547

Honors and Awards

- **China National Scholarship**
  top 0.2% | 12 2022
- **China National Scholarship**
  top 0.2% | 12 2025
- **Chinese Mathematics Competition for College Students**
  First Prize | top 5.3% | 12 2020
- **China International College Students' 'Internet+' Innovation and Entrepreneurship Competition**
  Silver Award | 0.02% | 11 2021

References

1. **Sai Duan**
   Young Researcher, Department of Chemistry
   Fudan University
   Email: duansai@fudan.edu.cn
2. **Xin Xu**
   Professor, Department of Chemistry
   Fudan University
   Email: xxchem@fudan.edu.cn